

Project 2: Data at Rest

Robert Simpson

University of Maryland Global Campus

DFCS 625: Windows Forensics

Randy Grizzelli

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Essay Question # 1

The Windows Shadow Copy Service, or VSS, manages how the Windows OS performs backups. VSS manages the communication between the data being backed up, the application performing the back up, and where the backup will be stored (Harwood, 2025). It utilizes four services to perform these functions. The first service, VSS Service, is simply the part of the Windows OS that manages communication. The VSS Requester is the backup application, and the VSS Provider creates and maintains the snapshot. Lastly, the VSS Writer ensures there is a consistent data set to backup to (Harwood, 2025). Essentially, VSS ensures that data can be backed up with as little chance of corruption or data loss possible.

For the digital forensic analyst, this is a potential gold mine. The service benefits the forensic analyst as it provides valuable and reliable data of what the computer was doing and what was on the computer at the given point in time (Ghafarian and Hills, 2018). If the investigation requires the examination of a suspect's laptop and certain data was deleted or corrupted, the analysts may be able to retrieve the data from the shadow copy. This is especially helpful if the backup was stored off-site. It should be noted that this may require a separate search warrant. The analysts could look at the data on the shadow copies and possibly recover the lost data. When multiple snapshots are pieced together, this can allow the analyst to build a timeline of events and build a more complete picture (Mohammed, 2023). This is dependent on how often the machine performs backups.

Essay Question # 2

In simple terms, BitLocker is Microsoft's proprietary drive encryption system for the Windows Operating System. Many people may be familiar with BitLocker if they ever restarted

a work computer and were presented with a blue screen asking for a BitLocker pin to unlock the computer. BitLocker encrypts the storage drive of the computer. When combined with a TPM at the hardware level, this acts as an anti-tamper mechanism (Officedocspr5, 2025). This means that if the drive is removed and put on another computer, it will not work. However, if the device does not have a TPM, BitLocker can be configured to use a startup key or password (Officedocspr5, 2025). This is less secure as it will not provide a pre-boot check. BitLocker has a variety of ways to both encrypt and decrypt drives. According to Microsoft, if the device has a TPM then BitLocker can encrypt with TPM only, TPM with startup key, TPM with a pin, or TPM with startup key and a pin (2025). The last will be the most secure. These additional methods act as multi-factor authentication, which adds an additional layer of security to the encrypted drive. Alternatively, to decrypt the drive, an authentication key or recovery key can be used or even BitLocker Network Unlock (Microsoft, 2025). In summary, BitLocker encrypts the storage drive on the computer. On top of that, a startup key stored on a USB or PIN can be configured to unlock the computer and storage drive. BitLocker can then decrypt the drive using a recovery or authentication key.

Essay Question # 3

Disk defragmentation is a process that is no longer needed if the machine is using SSDs rather than spinning disk HDDs. The reason for this is that HDDs have a physical spinning disk to write and store data. Over time, this data can become fragmented in different parts of the disk, leading to slower read times (Harwood, 2026). The process of disk defragmentation reorganizes the data to be closer together and thus have faster read times. However, SSDs do not have a physical spinning disk and instead store the data electronically. They have their own processes of handling the data and keeping it organized. Running disk defragmentation on an SSD could

potentially add far more write cycles to the drive than needed. This benefits the threat actor as SSDs only have so many write cycles available before the drive becomes degraded (DataRecovery, 2022). This can make the analyst's job more difficult as the drive could potentially become unusable.

There are alternative ways the threat actor can use native Windows tools to obscure their tracks. Cipher.exe could be used to perform write passes on the drive to overwrite the data with a series of 0's and random data (Gogolin, 2021). The Microsoft tool clear-disk could be used as well. Clear-disk, "Clears a disk by removing all partition information and un-initializing it, erasing all data on the disk" (Harwood, 2026). Although, if this is done and nothing else is written to the disk, this "deleted" data could still be recovered as was seen in DFCS 615 when the student analysts performed a similar task. However, this was performed manually and not with clear-disk. The analyst must understand that data is not simply "erased". In NTFS, the space in the Master File Table (MFT) will be marked as clear so that the system knows it can write data there. However, until the data is actually written, the previous data is still there.

Essay Question # 4

There are a few ways to decrypt a BitLocker drive depending on the information and tools at hand. The first, as discussed earlier, is to use the correct PIN, startup key, or recovery key. However, in many forensic scenarios, this information may not be available. The alternative is a much more complex and involved process. A study performed by Assumpcao et al. utilized Arsenal Image Mounter and Intel Direct Connect Interface (DCI) to obtain the Volume Master Key (VMK) and decrypt the drive (2023). The first part of the process uses the command "manage-bde" and its associated commands such as "manage-bde-status" to view the encryption status of the drive (Assumpcao et al., 2023). This is similar to what the analyst did during the

Unit 4 lab on the subject's workstation. After using Arsenal Image Mounter, the team physically connected to the machine using Intel DCI. The team then retrieved the Windows Boot Manager UEFI from the EFI system interface and ran it in Ghidra for reverse engineering (Assumpcao, 2023). Through this process, the team extracted the VMK and could then decrypt the drive (Assumpcao, 2023). With all this being said, it is clear that without the BitLocker PIN, startup key, or recovery key, it is a long and difficult process for the forensic analyst to decrypt the drive. This is the purpose of encryption. This makes it more difficult for anyone, not just analysts, to decrypt and potentially steal important data from the victim.

Essay Question # 5

Based on the sources presented thus far as well as anecdotal evidence from the unit 4 lab, the preferred method for a BitLocker image acquisition would be a live capture as long as the system is on. The unit 4 lab presented the analyst with a machine that was already running and logged into the user account. The steps taken to acquire the image in this scenario compared to the dead-box acquisition in the Assumpcao study were far simpler and less time consuming. A dead box acquisition requires either knowledge of the BitLocker PIN, recovery key, or the startup key (officedocspr5, 2025). If that knowledge is unknown than the analyst must resort to other hardware to extract the bits to obtain the VMK as seen in the Assumpcao study (2023). This is far more time consuming and more difficult compared to getting on the keyboard of a logged in system and running the appropriate commands as was performed in the lab.

Unit 4 Lab

As the security analyst, it was critical to gather data of the Subject's workstation and see if there were any encrypted drives. As seen in Figure 1, there were 2 fully decrypted volumes and one fully encrypted volume. The fully encrypted volume was Volume D:. I discovered this

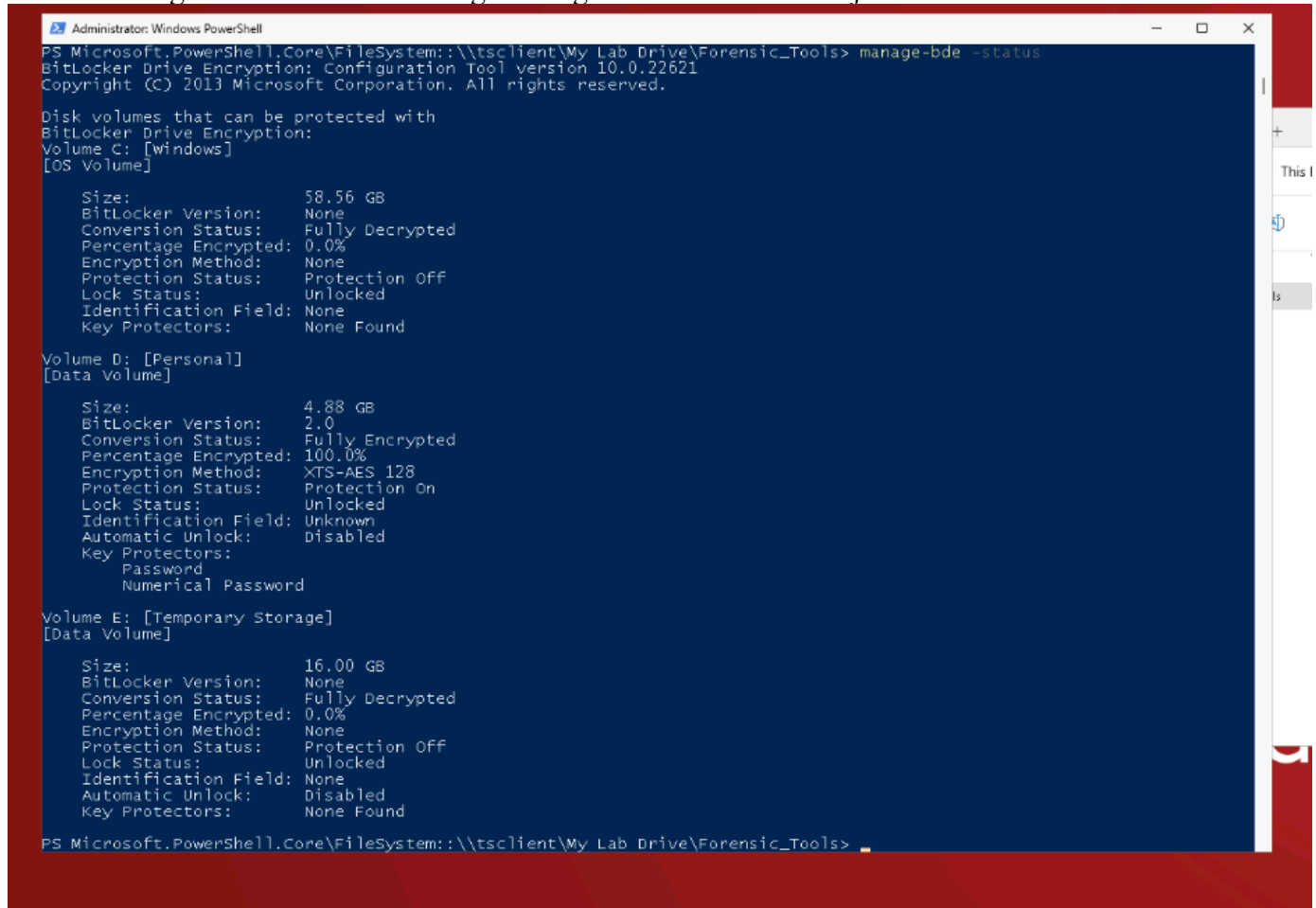
by first running PowerShell as administrator on the target workstation. I ran the command “manage-bde –status” and noticed that one volume was unknown. I then ran “manage-bde –unlock D: -password” and used the password “@dmin@dmin” to unlock it. Running “manage-bde –status” again showed that Volume D: was encrypted. Figure 2 shows a clearer screenshot of the volume. Two other volumes were present. The Windows OS was on Volume C: which used 58.56Gb and was fully decrypted since the OS was running. Temporary storage used 16Gb of Volume E. This also was fully decrypted, leaving only Volume D: as the encrypted volume. Running the command “manage-bde –protectors –get D:” provided me with the password and numerical password which can be seen in Figures 3 and 4. The first part of the output with “Password:” and “ID:” followed by a string of alphanumeric characters shows that there is a user configured password to decrypt the volume (officedocspr5, 2025). Figure 4 shows a clear screenshot of the numerical password which is the Recovery-Key and ID for BitLocker Volume D:. So, either the user generated password or the recovery key could be used to decrypt the volume. The ID for the password is: {B4A5548E-0359-4E9E-9F8B-73F584291CCF}. The ID for the numerical password is: {D8182F51-CF38-4598-ADF8-7CA118F287E6} and the recovery key is: {255002-593846-706090-485917-714142-335742-696905-501721}.

The next step in the process was to get a logical image of the encrypted Volume D:. FTK Imager was used for this. Once the image was completed, the SHA1 and MD5 hashes and the size of the image were noted from the verify results panel. The MD5 hash was: 90221b14477ac0cfd30472df2ae2eb4 and the SHA1 hash was: 4ecb3d1690dd3aef9e1efdec2f0ff06a05f913d. Additionally, the source data size was 5000MB or 5GB, of which 4.88GB was used as seen in figure 2. The hashes and size can be found in

figures 5 and 6. Figure 7 shows the exact Windows OS version the machine was running after running the “winver” command”. This was Windows version 23H2 OS Build 22631.3007. The final part of the process involved getting the tertiary evidence and data such as the IP configuration, network traffic, and running processes. Netstat was used to gather TCP connections. The command “netstat -ano” was specifically used. The “-a” gathers all of the TCP and UDP connections that the subject’s machine was listening on (Hardwood, 2024). The “-n” gathers IP addresses and port numbers from those TCP connections (Harwood, 2024). Lastly, the “-o” gets the process IDs or PIDS associated with those TCP connections (Harwood, 2024). This gives the analyst what idea of what the subject’s machine was in communication with such as potentially a server running over ports 443, 80, or 8080 with a known malicious IP address. Gathering process IDs can also help piece together potential malware.

- 1 Include a screenshot of the manage-bde -status output. This will verify you have correctly unlocked the drive and identified the encrypted volume.

Figure 1: Results of running “manage-bde –status” on Subject’s workstation



```
Administrator: Windows PowerShell
PS Microsoft.PowerShell.Core\FileSystem::\\tsclient\My Lab Drive\Forensic_Tools> manage-bde -status
BitLocker Drive Encryption: Configuration Tool version 10.0.22621
Copyright (C) 2013 Microsoft Corporation. All rights reserved.

Disk volumes that can be protected with
BitLocker Drive Encryption:
Volume C: [Windows]
[OS Volume]

    Size: 58.56 GB
    BitLocker Version: None
    Conversion Status: Fully Decrypted
    Percentage Encrypted: 0.0%
    Encryption Method: None
    Protection Status: Protection Off
    Lock Status: Unlocked
    Identification Field: None
    Key Protectors: None Found

Volume D: [Personal]
[Data Volume]

    Size: 4.88 GB
    BitLocker Version: 2.0
    Conversion Status: Fully Encrypted
    Percentage Encrypted: 100.0%
    Encryption Method: XTS-AES 128
    Protection Status: Protection On
    Lock Status: Unlocked
    Identification Field: Unknown
    Automatic Unlock: Disabled
    Key Protectors:
        Password
        Numerical Password

Volume E: [Temporary Storage]
[Data Volume]

    Size: 16.00 GB
    BitLocker Version: None
    Conversion Status: Fully Decrypted
    Percentage Encrypted: 0.0%
    Encryption Method: None
    Protection Status: Protection Off
    Lock Status: Unlocked
    Identification Field: None
    Automatic Unlock: Disabled
    Key Protectors: None Found

PS Microsoft.PowerShell.Core\FileSystem::\\tsclient\My Lab Drive\Forensic_Tools>
```

Figure 2: Zoomed in screenshot of “manage-bde –status” results on Volume D:

```
Volume D: [Personal]
[Data Volume]

Size: 4.88 GB
BitLocker Version: 2.0
Conversion Status: Fully Encrypted
Percentage Encrypted: 100.0%
Encryption Method: XTS-AES 128
Protection Status: Protection On
Lock Status: Unlocked
Identification Field: Unknown
Automatic Unlock: Disabled
Key Protectors:
    Password
    Numerical Password
```

- 2 Include a screenshot of the `manage-bde -get -protectors` output. This will verify you have correctly recovered the decryption key.

Figure 3: Passwords of Encrypted Volume

```
PS Microsoft.PowerShell.Core\FileSystem::\\tsclient\My Lab Drive\Forensic_Tools> manage-bde -protectors -get D: >> .\BitLocker_Protectors.txt
PS Microsoft.PowerShell.Core\FileSystem::\\tsclient\My Lab Drive\Forensic_Tools> manage-bde -protectors -get D:
BitLocker Drive Encryption: Configuration Tool version 10.0.22621
Copyright (C) 2013 Microsoft Corporation. All rights reserved.

Volume D: [Personal]
All Key Protectors

    Password:
        ID: {B4A5548E-0359-4E9E-9F8B-73F584291CCF}

    Numerical Password:
        ID: {D8182F51-CF38-4598-ADF8-7CA118F287E6}
        Password:
            255002-593846-706090-485914-714142-335742-696905-501721

PS Microsoft.PowerShell.Core\FileSystem::\\tsclient\My Lab Drive\Forensic_Tools> _
```

- 3 What is the Recovery-Key and ID for your BitLocker encrypted drive?

Figure 4: Zoomed in screenshot of Volume Recovery-Key and ID

```
Volume D: [Personal]
All Key Protectors

    Password:
        ID: {B4A5548E-0359-4E9E-9F8B-73F584291CCF}

    Numerical Password:
        ID: {D8182F51-CF38-4598-ADF8-7CA118F287E6}
        Password:
            255002-593846-706090-485914-714142-335742-696905-501721
```

- 4 What are the MD5 and SHA1 hash values for your BitLocker logical acquisition? What is the size of the logical acquisition?

Figure 5: MD5 and SHA1 hash of the logical image taken of the encrypted Volume D

```
WindowsTGTClient.001.txt - Notepad
File Edit Format View Help
Created By AccessData® FTK® Imager 4.7.2.11

Case Information:
Acquired using: ADI4.7.2.11
Case Number: WindowsTGTClient
Evidence Number: 1-1
Unique description: Encrypted Drive
Examiner: UMGC Student
Notes:

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Information for \\tsclient\My Lab Drive\WindowsTGTClient:

Physical Evidentiary Item (Source) Information:
[Device Info]
  Source Type: Logical
[Drive Geometry]
  Bytes per Sector: 512
  Sector Count: 10,240,000
[Physical Drive Information]
  Removable drive: False
  Source data size: 5000 MB
  Sector count: 10240000
[Computed Hashes]
  MD5 checksum: 90221b144776ac0cfd30472df2ae2eb4
  SHA1 checksum: 4ecb3d1690dd3aef9e1efdec2f0ff06a05f913d

Image Information:
Acquisition started: Fri May 29 14:06:56 2026
Acquisition finished: Fri May 29 14:27:34 2026
Segment list:
  \\tsclient\My Lab Drive\WindowsTGTClient.001

Image Verification Results:
Verification started: Fri May 29 14:27:35 2026
Verification finished: Fri May 29 14:36:57 2026
MD5 checksum: 90221b144776ac0cfd30472df2ae2eb4 : verified
SHA1 checksum: 4ecb3d1690dd3aef9e1efdec2f0ff06a05f913d : verified
```

Figure 6: Image size

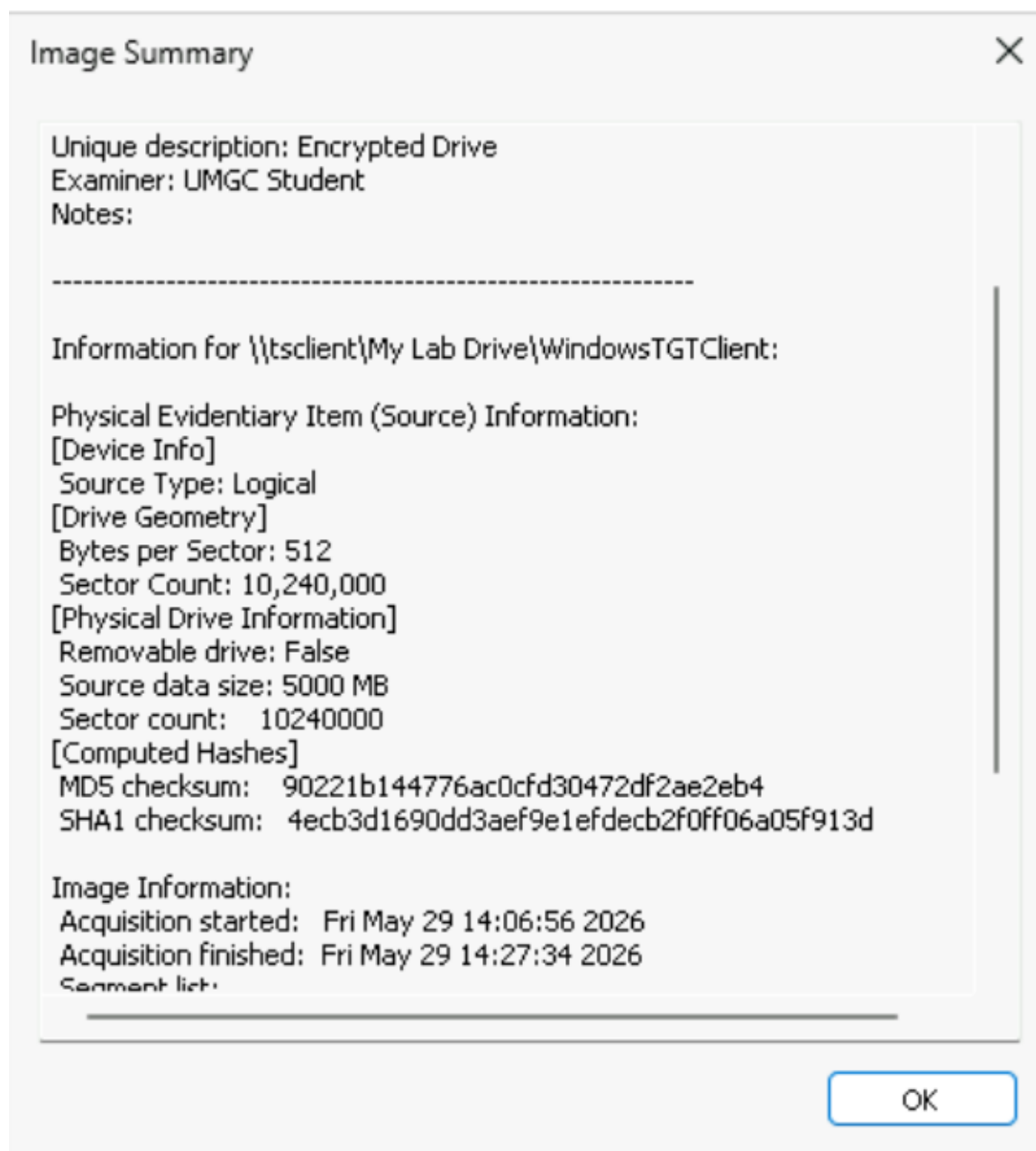


Figure 7: Windows version of Subject's workstation



Figure 8: Hash list of additional information gathered during investigation

	A	B	C	D	E
1	MD5.SHA1.FileNamees				
2	21412794e6e0b7202c3359cbe85386a1,"9d3131bba4479900c028ec11614cfde6421b653c",//tsclient/My	Lab	Drive\Wtsclient\My	Lab	Drive\Forensic_Collections\BitLocker_Protectors.txt"
3	bed5d118027c3ee8905a60726288d78,"b9fe1fa063bd903c79b4b150338d5402493cd95b",//tsclient/My	Lab	Drive\Wtsclient\My	Lab	Drive\Forensic_Collections\tasklist_dll.txt"
4	873340d4042cbd26f63c48fd3cab26a,"9f7edacd8c804a8d6e972faec718b80f0c2b2076",//tsclient/My	Lab	Drive\Wtsclient\My	Lab	Drive\Forensic_Collections\Connects\WithProcess.txt"
5	e6a8bb89e6077582c685338ce4b0edd1,"03d17c70d76148e3e94acedb8d5e01f6143c8d9",//tsclient/My	Lab	Drive\Wtsclient\My	Lab	Drive\Forensic_Collections\BitLocker_Status.txt"
6	091d46f9dcaa23ea8f891c5a33a76b0,"0394b262573e88d4db900db53ad65476379a67ac",//tsclient/My	Lab	Drive\Wtsclient\My	Lab	Drive\Forensic_Collections\inetstat.txt"
7	7e16acb822923e682efbcbdc629b60e,"da92c0091398ee6b49bedde79fa2e7b493664a29",//tsclient/My	Lab	Drive\Wtsclient\My	Lab	Drive\Forensic_Collections\tasklist.txt"
8	78d12818a164b6bc7006a88f548d62a4,"a780af1560e07e94b485a21542e0771945e33f7",//tsclient/My	Lab	Drive\Wtsclient\My	Lab	Drive\Forensic_Collections\ipconfig.txt"
9					

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